EE558 HW 3

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1. $P_{s,cout} = 26dBm$
   $P_{in} = 1.8dBm$ $\longrightarrow$ $G = P_{s,cout} - P_{in} = \underline{\underline{24.2dB}}$

2. 100 GHz channel spacing
   1536-1556nm spacing $\rightarrow$ 195.312-192.802 THz spacing = 2510 GHz spacing
   = 25 wavelength channels

3. EDFA amplifies all channels simultaneously:
   So for 1, 2, 3, and 8 wavelength channels:
   $P_{in} = 0.001mW = -30dBm$
   $P_{out} = -30dBm + 26dB = \underline{-4dBm}$

4. $\Delta\tau = D_{PMD}\sqrt{L} = 0.2ps/\sqrt{km} \times \sqrt{120km} = \underline{\underline{24ps}}$

5. Carrier frequency: 193.5THz $\rightarrow$ 1.5993 um, BW: 80GHz $\rightarrow$ 0.08THz
   upper range of BW: 193.58THz $\rightarrow$ 1.5487um
   $\Delta\lambda = 0.0006\,\mu m = 0.6nm$

   a) $\Delta\tau = 17ps/nm/km \times 0.6nm \times 80km = \underline{816ps}$

   b) $\Delta\tau = -80ps/nm/km \times 0.6nm \times 20km = \underline{-960ps}$

   c. for a: slower v (higher time delay) $\rightarrow v = \frac{c}{n} \rightarrow$ bigger $n \rightarrow$ shorter λ
   b: faster $v \longrightarrow v = \frac{c}{n} \rightarrow$ smaller $n \longrightarrow$ longer λ